

Assume that solution is of form

$$0 = \epsilon_1 = \dots = \epsilon_a < \epsilon_{a+1} \leq \dots \leq \epsilon_b < \epsilon_{b+1} = \dots = \epsilon_n$$

Quickselect based k-capped

General approach

Find a in linear time using quickselect

Find $b+1$ in linear time using quickselect

In constant time, find the $\frac{1}{2}\lambda$ to add or subtract from element j in the range

$$x_a \leq x_j \leq x_b$$

Step 1: a in linear time

Input: Original vector x

front $\leftarrow 1$

back $\leftarrow n$

while back - front ≥ 1

ind $\rightarrow$ partition(x, f, b) - 1

gamma = $-2 \times [\text{ind}]$ % Derived by setting $x_i + \frac{1}{2}\lambda = 0$

$$\text{sum} = \sum_{i=1}^n \max(0, \min(1, x_i + \frac{1}{2} \cdot \text{gamma}))$$

if sum $> k$

front $\rightarrow$ ind

else if sum $< k$

back $\rightarrow$ ind - 1

else if (sum = k) Return solution with calculated λ
return $\max(0, \min(1, x_i + \frac{1}{2} \cdot \text{gamma}))$ for every x_i in x

Step 2: $b+1$ in linear time

Input: Original vector x

front $\leftarrow 1$

back $\leftarrow n$

while back - front ≥ 1

ind $\rightarrow$ partition(x, f, b) - 1

gamma $\rightarrow -2 \times \text{ind} + 2$

sum $= \sum_{i=1}^n \max(0, \min(1, x_i + \frac{1}{2} \cdot \text{gamma}))$

if sum $> k$

~~back~~

back $\rightarrow$ ind - 1

else if sum $< k$

front $\rightarrow$ ind

else if (sum $= k$) Return solution with λ

return $\max(0, \min(1, x_i + \frac{1}{2} \cdot \text{gamma}))$

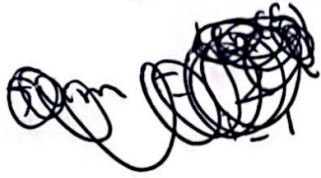
for every x_i in x

return front

$n+1 = \text{front}$

So far, I can prove things always work
unless $a = 0$ or ~~back~~ $b+1 \geq n+1$
(Edge cases we can deal with later)

Step 3: Determine Optimal gamma in constant time
 $\epsilon = []$
 for i in range(n)
 if $x[i] \geq x[a]$ and $x[i] \leq x[b]$
 $\epsilon.append(x[i])$



$$\text{half-gamma} = \frac{\text{sum}(\epsilon) - k}{\text{len}(\epsilon)}$$

$$\epsilon = \epsilon + \text{half-gamma}$$

return an array with "a" 0s, all the elements of ϵ and 1s such that the overall length is n

Optional verification step

make sure for all elements not added to ϵ , $x_i + \text{half-gamma} \leq 0$ or ≥ 1 .